

# Media Master: A Standalone Browser-Based Culture Media Preparation Calculator with an Integrated Microbiological Media Database

Ahmed Raslan<sup>1</sup>

<sup>1</sup>Department of Genetics, Faculty of Agriculture and Natural Resources,  
Aswan University, Aswan, Egypt

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## Abstract

The preparation of microbiological culture media is a routine yet error-prone task in bacteriology and mycology laboratories, requiring precise scaling of multi-component recipes to variable target volumes. Arithmetic errors during recipe scaling can lead to media with incorrect nutrient concentrations, inappropriate solidification, or selective agent imbalances, compromising experimental outcomes and wasting costly reagents. We present *Media Master*, an open-source, standalone HTML5 application that provides an integrated culture media preparation calculator coupled with a comprehensive database of 20 standard microbiological media formulations spanning fungal, general bacterial, selective, differential, and enriched categories. The application features multi-dimensional filtering by target organism (fungi, Gram-positive bacteria, Gram-negative bacteria) and media classification (general, selective, enriched, testing), real-time proportional recipe scaling for arbitrary volumes, optional antibiotic supplementation calculations, dual preparation method support for Potato Dextrose Agar (manual potato infusion vs. commercial powder), and a print-optimized recipe checklist with sterilization reminders. All computations are performed client-side using vanilla JavaScript, enabling full offline functionality with zero dependencies. Media Master is distributed as a single HTML file and operates on any device with a modern web browser, making it accessible to laboratories regardless of infrastructure constraints. This paper describes the software design, the embedded media database, the scaling algorithms, and the practical applications of Media Master in standardizing culture media preparation workflows.

**Keywords:** Culture media, Media preparation, Recipe scaling, Microbiology laboratory, Standalone software, HTML5, Open-source, Offline tools.

# 1 Introduction

The preparation of culture media is among the most fundamental and frequently performed procedures in any microbiology laboratory, underpinning virtually every aspect of experimental bacteriology and mycology from routine isolation and enumeration to antimicrobial susceptibility testing and metabolic profiling [1]. Standard microbiological media such as Nutrient Agar, Luria-Bertani (LB) Agar, Potato Dextrose Agar (PDA), Sabouraud Dextrose Agar (SDA), and MacConkey Agar are formulated as multi-component recipes typically standardized for 1-liter volumes, with each constituent specified in grams per liter (g/L) [2]. In practice, however, researchers and laboratory technicians rarely require exactly one liter of prepared media. Depending on the experimental design, the number of Petri dishes to be poured, or the available autoclave capacity, the target volume may range from as little as 100 mL to several liters. This necessitates proportional scaling of every ingredient in the recipe—a seemingly trivial arithmetic exercise that, when performed manually under the time pressures and cognitive load of active bench work, becomes a surprisingly common source of error [4].

The consequences of recipe scaling errors in culture media preparation extend far beyond mere inconvenience. An incorrect concentration of agar can produce media that fails to solidify properly or forms an excessively rigid gel that inhibits colony spreading and morphological assessment. Errors in the concentration of selective agents—such as bile salts in MacConkey Agar, crystal violet in Eosin Methylene Blue (EMB) Agar, or sodium chloride in Mannitol Salt Agar (MSA)—can fundamentally alter the selective and differential properties of the medium, leading to false-negative isolation of target organisms or false-positive growth of non-target contaminants [8]. In antifungal or antibacterial screening studies, where media may be supplemented with precise concentrations of antimicrobial compounds, even small deviations in the base media composition can confound the interpretation of minimum inhibitory concentrations (MICs) and zone diameters. These errors are particularly insidious because they may not be immediately apparent, potentially corrupting entire experimental datasets before being detected.

Despite the critical importance of accurate media preparation, the computational tools available to laboratory personnel for this task remain remarkably primitive. Most laboratories rely on manual calculations performed on paper, basic calculator applications, or generic spreadsheet templates that must be individually constructed and maintained for each media type [7]. Dedicated media preparation software does exist in the form of proprietary laboratory information management systems (LIMS), but these platforms are typically designed for industrial-scale operations and impose substantial costs in terms of licensing, IT infrastructure, and training [5]. Cloud-based calculators, while accessible, introduce dependencies on internet connectivity that render them unusable during network outages—a particularly problematic limitation in laboratories located in regions with unreliable infrastructure [6]. Furthermore, existing calculator tools rarely integrate a comprehensive media database, requiring the user to manually look up and enter ingredient compositions from textbooks or manufacturer datasheets before performing the scaling calculation.

To address this gap, we developed *Media Master*, an open-source, standalone HTML5 application that combines a curated database of 20 commonly used microbiological media formulations with an automated, real-time recipe scaling calculator. The application features multi-dimensional filtering by target organism and media classification, optional antibiotic supplementation calculations, dual preparation method support for traditional

and commercial PDA formulations, and a print-optimized recipe checklist with integrated sterilization reminders. By encapsulating both the reference database and the computational engine within a single, dependency-free HTML file, Media Master eliminates the need for external references, internet connectivity, or specialized software, providing laboratories of all scales and resource levels with a reliable, portable tool for standardizing their media preparation workflows.

## 2 Software Architecture

### 2.1 Client-Side Standalone Design

Media Master is engineered as a fully self-contained, client-side web application following the single-page application (SPA) paradigm [3]. The complete application—including the user interface markup (HTML5), styling rules (Tailwind CSS), the embedded media database, and all computational logic (vanilla JavaScript ES6+)—is encapsulated within a single `.html` file. This monolithic file can be executed directly in any modern web browser by opening it from the local filesystem via the `file://` protocol, without requiring a web server, runtime environment, package manager, or active internet connection. The single-file distribution model enables the tool to be shared through the most basic channels available in resource-limited settings, including USB flash drives, email attachments, or institutional intranets.

### 2.2 User Interface Design

The user interface is constructed using Tailwind CSS with a refined typographic system employing the DM Serif Display font for headings and Inter for body text, establishing a professional aesthetic appropriate for a laboratory reference tool [10]. The layout is responsive, adapting seamlessly to desktop monitors, tablets, and mobile devices through CSS grid breakpoints. Key interaction elements include dropdown selectors with real-time filtering, radio button toggles for preparation method selection, and checkbox controls for additive supplementation. All user inputs trigger immediate recipe recalculation through JavaScript event listeners, providing continuous visual feedback without requiring explicit submission actions.

### 2.3 Data Architecture

The media database is implemented as a structured JavaScript object literal embedded directly within the application source code. Each media entry contains the following fields: a human-readable display name, a functional category (General, Selective, Enriched, or Testing), a target organism classification (Fungi, Bacteria, Gram-positive, or Gram-negative), a descriptive text summarizing the medium’s purpose and applications, a source URL linking to the manufacturer’s datasheet or authoritative reference, and one or more recipe variants specifying the ingredient name, rate (g/L), and unit for each component. This self-contained data architecture eliminates external API dependencies and ensures that the entire knowledge base remains accessible offline.

## 2.4 Privacy and Data Handling

In contrast to cloud-based calculators that may log usage data or require user accounts, Media Master performs all operations exclusively within the browser’s JavaScript runtime. No data is transmitted over any network, no cookies are set, and no persistent storage is utilized. The application’s computational state exists transiently in the browser’s Document Object Model (DOM) and is discarded entirely upon page closure. This ephemeral data model ensures complete operational privacy, a relevant consideration for laboratories operating under institutional data governance policies [9].

## 3 Integrated Media Database

Media Master incorporates a curated database of 20 standard microbiological media formulations, organized into four functional categories across two target organism domains. Table 1 provides a comprehensive listing of all included media with their classifications and primary applications. Each entry is linked to authoritative source references including manufacturer datasheets (Sigma-Aldrich, Neogen) and peer-reviewed microbiological resources.

Table 1: Complete listing of culture media included in the Media Master database, organized by target organism and functional category.

| Abbreviation                                  | Target        | Category  | Primary Application                                                  |
|-----------------------------------------------|---------------|-----------|----------------------------------------------------------------------|
| <b>Fungal Media</b>                           |               |           |                                                                      |
| PDA                                           | Fungi         | General   | Cultivation and enumeration of yeasts and molds                      |
| PDB                                           | Fungi         | General   | Liquid cultivation and biomass production                            |
| MEA                                           | Fungi         | General   | Isolation and maintenance of soil and wood fungi                     |
| SDA                                           | Fungi         | Selective | Cultivation of dermatophytes and pathogenic fungi                    |
| CZA                                           | Fungi         | General   | Taxonomic studies of <i>Aspergillus</i> and <i>Penicillium</i>       |
| YPD                                           | Fungi         | Enriched  | Routine growth of <i>Saccharomyces cerevisiae</i>                    |
| DRBC                                          | Fungi         | Selective | Selective enumeration of molds in food samples                       |
| <b>General/Enriched Bacterial Media</b>       |               |           |                                                                      |
| NA                                            | Bacteria      | General   | General cultivation of non-fastidious organisms                      |
| LB                                            | Bacteria      | General   | <i>E. coli</i> cultivation in molecular biology                      |
| TSA                                           | Bacteria      | General   | General-purpose for fastidious and non-fastidious bacteria           |
| BHI                                           | Bacteria      | Enriched  | Fastidious pathogenic bacteria, yeasts, and molds                    |
| BAB                                           | Bacteria      | Enriched  | Blood agar base for hemolytic reaction observation                   |
| MHA                                           | Bacteria      | Testing   | Antimicrobial susceptibility testing (Kirby-Bauer)                   |
| <b>Selective/Differential Bacterial Media</b> |               |           |                                                                      |
| MAC                                           | Gram-negative | Selective | Differentiation of lactose-fermenting enterics                       |
| EMB                                           | Gram-negative | Selective | Isolation of Gram-negative enterics ( <i>E. coli</i> metallic sheen) |
| MSA                                           | Gram-positive | Selective | Isolation of pathogenic staphylococci                                |
| TCBS                                          | Gram-negative | Selective | Selective isolation of <i>Vibrio</i> species                         |
| SS                                            | Gram-negative | Selective | Isolation of <i>Salmonella</i> and <i>Shigella</i>                   |
| XLD                                           | Gram-negative | Selective | Differentiation of enteric pathogens                                 |
| BPA                                           | Gram-positive | Selective | Enumeration of coagulase-positive staphylococci                      |

### 3.1 Multi-Dimensional Filtering System

The database is accessible through a three-axis filtering system that allows users to narrow the media selection based on practical laboratory requirements:

1. **Target Organism Filter:** Restricts the displayed media to those designed for fungi and yeasts, all bacteria, Gram-negative bacteria only, or Gram-positive bacteria only.
2. **Category Filter:** Restricts selection to general/universal media, selective/differential media, enriched media, or testing/assay media.
3. **Text Search:** A free-text search field that performs real-time substring matching against media names, enabling rapid location of specific formulations.

These filters operate conjunctively (AND logic), allowing the user to execute highly specific queries such as “selective media for Gram-negative bacteria” with a single interaction. The filtered results dynamically populate the media selection dropdown, with a visual indicator displayed when no media match the current filter combination.

### 3.2 Recipe Composition Data

Each media entry in the database stores its recipe as an array of ingredient objects, where each object specifies:

- **name:** The chemical or common name of the ingredient (e.g., “Peptone,” “NaCl,” “Agar”).
- **rate:** The mass required per 1000 mL (grams per liter), serving as the base rate for proportional scaling.
- **unit:** The measurement unit (uniformly “g” for mass-based ingredients).

For Potato Dextrose Agar (PDA), two distinct recipe variants are stored: a “manual” variant specifying raw potatoes, dextrose, and agar for traditional preparation from fresh potato infusion, and a “commercial” variant specifying the mass of dehydrated commercial PDA powder. This dual-variant system reflects the practical reality that many laboratories, particularly those in resource-limited settings, prepare PDA from fresh potatoes rather than commercial powder due to cost or supply chain constraints.

## 4 Features and Computational Methods

### 4.1 Proportional Recipe Scaling

The core computational function of Media Master is the proportional scaling of media recipes from their standard 1-liter reference formulations to any user-specified target volume. For each ingredient  $i$  with a reference rate  $r_i$  (g/L) and a target volume  $V$  (mL), the required mass  $m_i$  is calculated as:

$$m_i = r_i \times \frac{V}{1000} \quad (1)$$

This calculation is performed in real-time as the user modifies the target volume input, with results displayed to two decimal places to accommodate the precision requirements of sensitive ingredients such as indicator dyes (e.g., Neutral Red at 0.03 g/L) or selective agents (e.g., Brilliant Green at 0.0003 g/L). The scaling algorithm handles all ingredients uniformly, ensuring that the proportional relationships between components are maintained regardless of the target volume.

## 4.2 Antibiotic Supplementation Calculator

Media Master includes an optional antibiotic supplementation module for calculating the mass of chloramphenicol (or other heat-labile antibiotics) required to achieve a specified concentration in the final media volume. When the user enables antibiotic supplementation and specifies the desired concentration in parts per million (ppm), the required mass is calculated as:

$$m_{\text{antibiotic}} = \frac{C_{\text{ppm}} \times V_{\text{mL}}}{1000 \times 1000} \text{ g} \quad (2)$$

where  $C_{\text{ppm}}$  is the target concentration in mg/L (equivalent to ppm for aqueous solutions) and  $V_{\text{mL}}$  is the total media volume. A prominent safety warning is automatically displayed when antibiotic supplementation is enabled, reminding the user that antibiotics must be added *after* autoclaving when the media has cooled to approximately 50°C to prevent thermal degradation of the compound.

## 4.3 Dual PDA Preparation Methods

Potato Dextrose Agar occupies a unique position among microbiological media in that it can be prepared either from commercial dehydrated powder or from fresh potatoes using a traditional boiling and infusion procedure [1]. Media Master supports both preparation methods through a toggle interface that dynamically switches between the two recipe variants:

- **Manual Method:** Displays the quantities of sliced unpeeled potatoes (200 g/L), dextrose (20 g/L), and agar (15 g/L) required, along with a step-by-step preparation guide including boiling volume, filtration instructions, and volume adjustment.
- **Commercial Method:** Displays the mass of commercial PDA powder (39 g/L) required for the specified volume.

This dual-method support acknowledges the diverse preparation practices across laboratories and educational settings, providing accurate calculations for both approaches within a single interface.

## 4.4 Recipe Checklist and Sterilization Guidance

Upon recipe calculation, the results are presented as a visually structured checklist that mirrors the sequential workflow of media preparation. Each ingredient is displayed with its calculated mass, formatted for easy reading during weighing operations. The checklist includes:

- Ingredient name and calculated mass with appropriate unit precision.
- Visual checkboxes for tracking weighing progress.
- Contextual color coding distinguishing standard ingredients from additives and special components.
- Estimated preparation time based on the media type.
- Standard autoclaving parameters (121°C for 15 minutes at 15 psi).
- Conditional safety alerts for heat-labile additives.

## 4.5 Media Information System

When a medium is selected, Media Master displays a detailed information card containing the full media name, descriptive badges indicating the target organism and functional category, a concise summary of the medium’s primary applications and selectivity principles, and a clickable hyperlink to the original source reference (e.g., manufacturer datasheet or authoritative protocol). This integrated reference system eliminates the need for users to consult external textbooks or online resources to verify media purpose and composition, consolidating the informational and computational aspects of media preparation within a single interface.

## 4.6 Output and Sharing

Media Master provides two output mechanisms:

1. **Print Recipe:** Invokes the browser’s native print dialog, rendering the recipe checklist in a clean, printer-friendly format suitable for posting at the media preparation bench.
2. **Copy to Clipboard:** Generates a plain-text summary of the recipe (media name, volume, and all ingredient masses) and copies it to the system clipboard for pasting into electronic laboratory notebooks, messaging applications, or email.

## 5 Conclusion and Future Directions

We have presented Media Master, an open-source, standalone HTML5 application that addresses the persistent need for accurate, accessible, and portable culture media preparation tools in microbiology laboratories. By integrating a curated database of 20 standard microbiological media formulations with automated proportional scaling, antibiotic supplementation calculations, and contextual preparation guidance, Media Master consolidates what has traditionally been a fragmented workflow involving textbooks, calculators, and handwritten notes into a single, unified interface.

The application’s standalone architecture—requiring no installation, server infrastructure, or internet connectivity—ensures that it is equally functional in well-equipped institutional laboratories and in resource-limited field or teaching settings. The single-file distribution model allows the tool to reach users through the simplest available channels, and the browser-based execution environment guarantees cross-platform compatibility without version-specific dependencies. The dual PDA preparation method support reflects a practical awareness of the diverse media preparation practices encountered across different laboratory contexts, from well-funded research institutions with access to commercial dehydrated media to teaching laboratories and field stations that may rely on traditional preparation from raw ingredients.

Several directions for future development are envisioned. First, the media database will be expanded to include additional specialized formulations such as chromogenic media, anaerobic culture media, and mycobacterial growth media, with the goal of providing comprehensive coverage of the media types encountered in clinical, environmental, and industrial microbiology. Second, a custom media builder will be implemented, allowing users to define, save, and recall proprietary or non-standard formulations tailored to their specific research needs. Third, a batch preparation mode will be developed to calculate ingredient requirements for multiple media types simultaneously, supporting the workflow

of laboratories that routinely prepare diverse media sets in a single session. Fourth, a pH adjustment calculator will be integrated to assist with the precise buffering and titration steps required for media such as Czapek Dox Agar and SDA that specify target pH values. Finally, the application will be extended with `localStorage`-based persistence to maintain a history of recent preparations, enabling users to quickly repeat commonly used recipes without re-entering parameters.

## Software Availability

Media Master is released as open-source software under the MIT License. The complete source code is available as a single HTML file and can be accessed at the project repository. No installation, registration, or internet connection is required for operation.



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